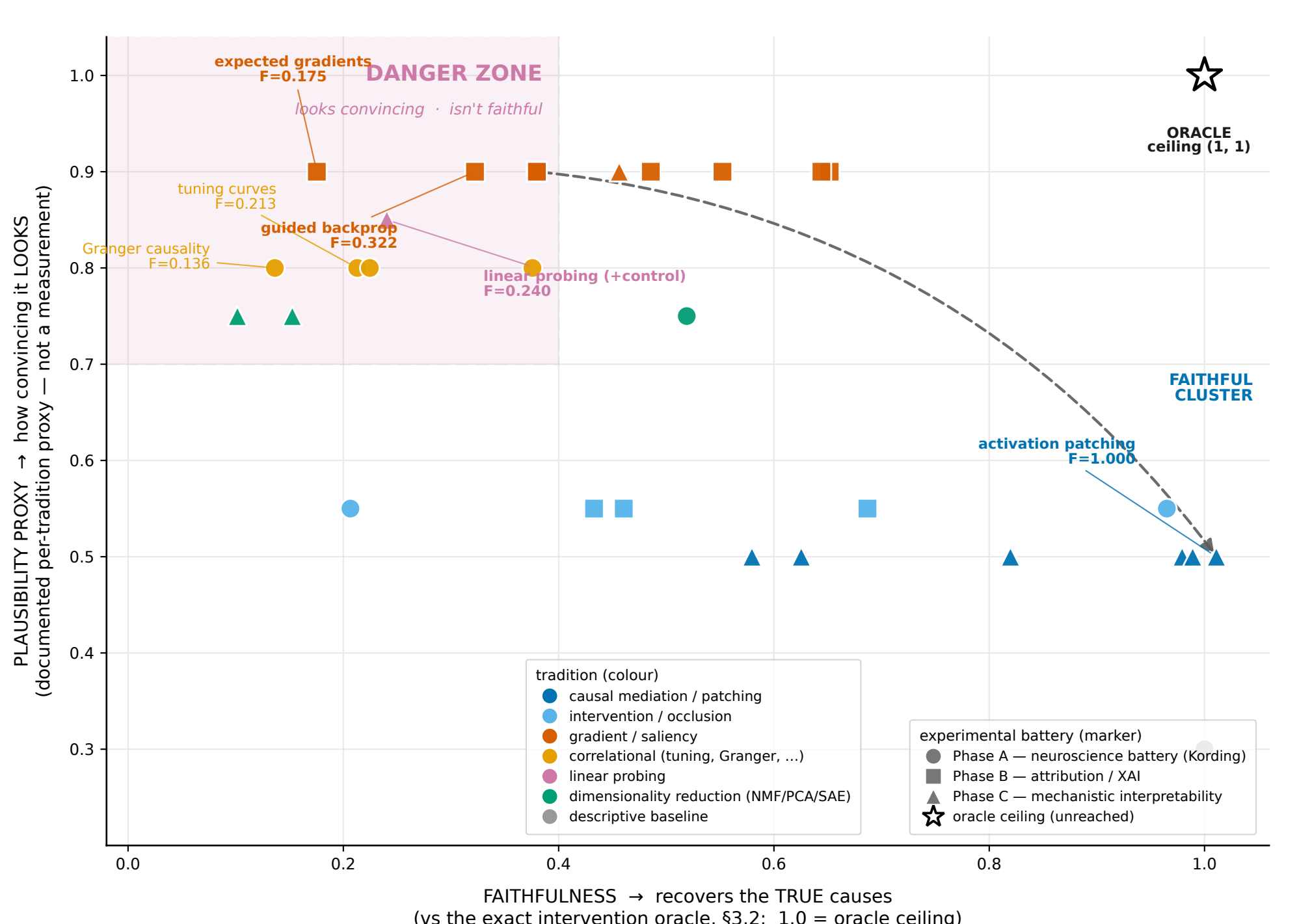

DANGER ZONE
looks convincing · isn't faithful
expected gradients
F=0.175
tuning curves
F=0.213
guided backprop
F=0.322
Granger causality
F=0.136
linear probing (+control)
F=0.240
ORACLE
ceiling (1, 1)
FAITHFUL
CLUSTER
activation patching
F=1.000
PLAUSIBILITY PROXY → how convincing it LOOKS
(documented per-tradition proxy — not a measurement)
FAITHFULNESS → recovers the TRUE causes
(vs the exact intervention oracle, §3.2; 1.0 = oracle ceiling)
tradition (colour)
causal mediation / patching
intervention / occlusion
gradient / saliency
correlational (tuning, Granger, ...)
linear probing
dimensionality reduction (NMF/PCA/SAE)
descriptive baseline
experimental battery (marker)
Phase A — neuroscience battery (Kording)
Phase B — attribution / XAI
Phase C — mechanistic interpretability
oracle ceiling (unreached)